

Decision Tree Inducers

Prerequisites

Before proceeding through this notebook, students should have completed

- Decision Tree basics
- Impurity Metrics

Learning Objectives

After going through this notebook, students should be able to:

- List popular decision tree inducers.
- List differences between ID3, C4.5, and CART.
- Use algorithms discussed in this chapter and python to create decision tree from scratch.

Introduction

In previous notebooks we studied about impurity metrics and pruning methods. In this notebook we will see how different impurity metrics and pruning methods are used by decision tree inducers.

Decision Tree inducers are algorithms used to create decision tree from the training data. Some popular decision tree inducers are:

- ID3
- C4.5
- CART

ID3

ID3 stands for Iterative Dichotomiser 3. 'Iterate' means to perform repeatedly, and 'dichotomize' means to represent as divided. ID3 repeatedly divides data at each node and hence the name. It was developed by Ross Quinlan and is used to solve classification problems. It creates multiway splits. For an n -valued categorical attribute, ID3 makes n splits. For example, if an attribute **outlook** has three values *sunny*, *foggy*, and *rainy*, then ID3 makes three splits, one for each distinct value of **outlook**.

ID3 uses **information gain to determine the best split**. The tree growing process stops when all instances are assigned a class label or when maximum information gain is not greater than 0. ID3 **doesn't use any tree pruning method**.

Algorithm

1. For all instances in a dataset D

If all instances belong to the same class C , or other stopping criteria(pure instances, maximum tree depth) are met

Create a leaf node and stop.

Else

Compute information gain for all previously unselected attributes.

Select an attribute(say f) that yields the greatest information gain.

if $max(\text{information gain}) < 0$

Create a leaf node and stop.

else

Split the data into subsets according to the value of attribute f .

2.Apply the algorithm recursively from step-1 for each of the subsets.

One major disadvantage of ID3 is that it is prone to overfitting because it doesn't use pruning method. Another disadvantage of ID3 is that it can only handle categorical attributes.

C4.5

C4.5, also developed by Ross Quinlan, is an improved version of ID3. Like ID3, C4.5 creates multiway splits for a categorical attribute and is used to solve classification problems.

Unlike ID3, C4.5 removed the restriction that attributes must be categorical. For continuous attributes, C4.5 creates a threshold and then splits the data into two parts. The first part contains all instances for which the attribute value is less than or equal to the threshold. The second part contains all instances for which the attribute value is greater than the threshold. It uses **gain ratio as splitting** criteria. The tree growing process stops when the number of instances to be split is below a certain threshold. For pruning, C4.5 **uses Error-based pruning** to solve the overfitting problem. Our previous notebook on pruning methods doesn't discuss error-based pruning, but you may refer to the additional resources section if you want to know more about error-based pruning.

Algorithm

1.For all instances in a dataset D

If all instances belong to the same class C , or other stopping criteria(pure instances, maximum tree depth) are met

Create a leaf node and stop

Else

- Compute gain ratio for all attributes.
 - Select an attribute(say f) that yields the greatest gain ratio.
 - Split the data into subsets according to the value of attribute f .
- 2. Apply algorithm recursively from step-1 for each of the subset.
- 3. Prune tree using error based pruning.

CART

CART stands for **Classification and Regression Trees**. CART constructs binary trees. Each split generates maximum of two subsets. For example, if an attribute **outlook** has 3 values *sunny*, *foggy*, and *rainy* then one subset may contain all instances in which **outlook** has value *foggy*. The other subset may contain instances in which **outlook** has value *sunny* and *rainy*. The splits are selected using **information gain** as the impurity metric and the tree is pruned using **cost-complexity pruning**.

An advantage of CART over ID3 and C4.5 is that it can be used for both regression and classification.

Algorithm(Classification)

1. For all instances in a dataset D
 - If all instances belong to the same class C , or other stopping criteria(pure instances, maximum tree depth) are met
 - Create a leaf node and stop
 - Else
 - Compute information gain for all attributes.
 - Select an attribute(say f) that yields the greatest information gain.
 - Split the data into subsets according to the value of attribute f .
2. Apply algorithm recursively from step-1 for each of the subset.
3. Prune tree using cost complexity pruning.

The algorithm uses information gain as the impurity metric but we could also use twoing criteria(Refer additional resources section for details).

Algorithm(Regression)

The above algorithm presents steps to construct a decision tree for classification problem using CART. Since, CART can be used for both classification and regression, let us also discuss the steps involved in creating regression trees.

1. For all instances in a dataset D
 - If all instances belong to the same class C , or other stopping criteria(MSE less than some threshold value) are met

```

    Create a leaf node and stop
Else
    Compute MSE for all attributes.
    Select an attribute(say  $f$ ) that yields the least MSE.
    Split the data into subsets according to the value of attribute  $f$ .

2.Apply algorithm recursively from step-1 for each of the subset.

3.Prune tree using cost complexity pruning.

```

The algorithm uses MSE as the impurity metric but we could also use variance or Residual Sum of Squares(RSS).

Now that we have studied decision tree inducers, we can conclude that among ID3, C4.5, and CART the easiest to implement is ID3. However, since CART can be used for both regression and classification, it is the most versatile. Due to this reason CART is used by `Scikit-learn` to create `DecisionTreeClassifier` and `DecisionTreeRegressor`.

The following table summarizes above algorithms:

Characteristics→	Impurity Metric	Attribute Type	Pruning Method	Application
Algorithm ↓				
ID3	Information gain	Handles only categorical attributes	None	Only used for classification
C4.5	Gain ratio	Handles both categorical and numerical attributes	Error Based Pruning	Only used for classification
CART	Information gain and twoing criteria for classification and MSE, RSS, and variance for regression	Handles both categorical and numerical attributes	Cost Complexity Pruning	Used for classification and regression

Additional Resources

Decision Tree Inducers:

- Lior Rokach, and Oded Maimon, [Decision Trees](#)
 - See Section 8 page 181 to read about Decision Tree Inducers

Pruning Method:

- Lior Rokach, and Oded Maimon, [Decision Trees](#)
 - See Section 3.11 page 172 to read about Twoing Criteria
 - See Section 6.4 page 176 to read about Minimum Error Pruning
- John Mingers, [An Empirical Comparison of Pruning Methods for Decision Tree Induction](#)
 - See section 2.2.3 to read about Minimum Error Pruning
 - See section 2.2.4 to read about Reduced Error Pruning

